Model Evaluation - VisionGATE Questions

GATE Data Science and Artificial Intelligence (GATE DA)

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Document compiled for GATE DA Preparation

1 Bias-Variance

Q1. Irreducible error arises from:

(A) using too simple a hypothesis class

(B) sensitivity of the learned model to the particular training sample

(C) inherent noise in the target — variability in y that no function of x can explain

(D) numerical errors in the optimizer

Q2. The bias of a learning algorithm at a point x is:

(A) the error of the single model you happened to train

(B) the difference between the *average* prediction over many training sets and the true $f(x)$

(C) the spread of predictions across training sets

(D) validation error minus training error

Q3. Select all true statements. (Select all that apply)

(A) Variance is an expectation over the random draw of the training set, not the test point

(B) Irreducible error shrinks as you collect more training data

(C) A model with zero variance must also have zero bias

(D) Both bias and variance are defined by averaging over training sets

(E) Irreducible error lower-bounds the expected test error

For Q4–Q7, use $\mathbb{E}\left[(y - \hat{f}(x))^2\right] = \text{Bias}^2 + \text{Var} + \sigma^2$.

Q4. Expected squared test error $= 0.50$, $\text{Bias}^2 = 0.20$, $\sigma^2 = 0.05$. Variance $=$

(A) 0.25

(B) 0.30

(C) 0.45

(D) 0.75

Q5. Bias $= 0.3$, Variance $= 0.16$, $\sigma^2 = 0.04$. Expected squared error $=$

(A) 0.29

(B) 0.50

(C) 0.25

(D) 0.20

Q6. Model A: $\text{Bias}^2 = 0.30$, Var $= 0.05$. Model B: $\text{Bias}^2 = 0.05$, Var $= 0.35$. Same $\sigma^2 = 0.10$. Which generalizes better?

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(A) Model A
(B) Model B
(C) Identical
(D) Cannot be determined

Q7. Raising the regularization strength moves variance $0.20 \to 0.08$ and Bias^2 from $0.05 \to 0.12$, with $\sigma^2 = 0.03$ fixed. Select all true statements. (Select all that apply)

(A) Total error fell from 0.28 to 0.23
(B) The trade was worthwhile
(C) The irreducible error also fell
(D) Increasing λ further is guaranteed to help

Q8. Darts tightly clustered, but the cluster sits well off the bullseye:

(A) Low bias, low variance
(B) Low bias, high variance
(C) High bias, low variance
(D) High bias, high variance

Q9. Darts scattered all over the board, but their centroid lands on the bullseye:

(A) Low bias, low variance
(B) Low bias, high variance
(C) High bias, low variance
(D) High bias, high variance

Q10. Which models plausibly produce the *scattered-but-centred* board? (Select all that apply)

(A) 1-nearest neighbour
(B) A fully grown, unpruned decision tree
(C) Predicting the global mean of y for every x
(D) A degree-15 polynomial fit to 20 points
(E) Ridge regression with very large λ

Q11. As model capacity increases, the typical pattern is:

(A) Bias $\uparrow$, variance $\downarrow$, test error $\downarrow$ monotonically
(B) Bias $\downarrow$, variance $\uparrow$, test error U-shaped
(C) Bias $\downarrow$, variance $\downarrow$, test error $\downarrow$
(D) Bias $\uparrow$, variance $\uparrow$, test error $\uparrow$

Q12. The minimum of the test-error curve occurs where:

(A) Training error first reaches zero
(B) Bias equals variance exactly
(C) The marginal decrease in Bias^2 equals the marginal increase in variance
(D) Capacity equals the number of training points

Q13. Select all true statements about the standard capacity curves. (Select all that apply)

(A) Training error is non-increasing in capacity
(B) The train–test gap tends to widen with capacity
(C) The test-error curve cannot fall below σ^2
(D) Variance decreases as capacity grows

Q14. Train MSE $= 0.02$, validation MSE $= 0.35$, noise floor ≈ 0.02. The model:

(A) Underfits — high bias
(B) Overfits — high variance
(C) Is well-balanced
(D) Has high irreducible error

Q15. Train error $= 0.24$, validation error $= 0.26$, noise floor ≈ 0.02. Best next step:

(A) Collect more training data
(B) Add regularization
(C) Increase model capacity / add features
(D) Reduce the number of features

Q16. Train error $= 0.20$, validation error $= 0.45$, noise floor ≈ 0.02. Select all true statements. (Select all that apply)

(A) Avoidable bias is roughly 0.18
(B) The variance gap is roughly 0.25
(C) The model suffers from both high bias and high variance
(D) Shrinking the model alone will fix it

Q17. With the hypothesis class held fixed, adding more training data:

(A) Reduces variance; bias essentially unchanged
(B) Reduces bias; variance unchanged
(C) Reduces both bias and irreducible error

(D) Increases variance because there is more noise in total

Q18. You add many extra features, some informative and some pure noise. Select all true statements. (Select all that apply)

(A) Bias tends to decrease

(B) Variance tends to increase

(C) Irreducible error decreases

(D) Overfitting risk rises, especially when d approaches n

Q19. Which of the following statements are True? (Check all that apply)

(A) Adding more training examples is a reliable fix for a high-variance problem

(B) A learning curve where training and cross-validation error converge to a high value indicates high variance

(C) Irreducible error can be reduced by choosing a more flexible hypothesis class

(D) If training error is low and cross-validation error is much higher, the model is likely overfitting

Q20. You plot learning curves and find that training error and cross-validation error have both flattened out, are close to each other, and both sit well above the desired performance level. Adding more training data will most likely:

(A) Substantially reduce test error

(B) Have little effect on test error

(C) Increase training error and decrease test error sharply

(D) Reduce the irreducible error

Q21. Suppose you have trained a regularized logistic regression model. It achieves 2% error on the training set and 22% error on the cross-validation set. Which of the following steps are likely to help? (Select all that apply)

(A) Increase the regularization parameter λ

(B) Decrease the regularization parameter λ

(C) Get more training examples

(D) Add more polynomial features

Q22. In curve fitting, you increase the degree of the polynomial from 2 to 12 while keeping the training set size fixed. Which of the following statements are True? (Select all that apply)

(A) Training error will typically decrease

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(B) Bias will typically decrease

(C) Variance will typically decrease

(D) Test error may first decrease and then increase

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2 Validation

Q1. When using a simple holdout validation approach (e.g. a single 70/30 train/test split) on a very small dataset, what is the most significant statistical risk?

(A) The model is guaranteed to overfit to the test set

(B) The evaluation metric will have high variance depending entirely on which data points happen to end up in the test set

(C) The training time will be prohibitively high

(D) Data leakage will naturally occur between the splits

Q2. Which of the following statements about train/validation/test split ratios are TRUE? (Select all that apply)

(A) A 50/50 split is globally considered the optimal choice to ensure equal statistical representation

(B) For massive datasets (e.g. 10 million rows), a 98/1/1 (train/val/test) split is often sufficient because 1% still yields enough rows for a robust evaluation

(C) Allocating too little data to the test set can lead to a high-variance estimate of model performance

(D) The validation set is meant for final, unbiased reporting of the model's accuracy to stakeholders

Q3. For a dataset containing N observations, Leave-One-Out Cross-Validation (LOOCV) requires fitting the model from scratch how many times?

(A) $N - 1$

(B) N

(C) N^2

(D) k

Q4. Which of the following are TRUE characteristics of Leave-One-Out Cross-Validation (LOOCV)? (Select all that apply)

(A) It produces completely deterministic results (there is no randomness in how the data is split)

(B) It is highly computationally efficient for deep neural networks on large datasets

(C) The models trained in each iteration are highly correlated with one another because their training sets are nearly identical

(D) It usually yields a high-bias estimate of the test error

Q5. If you are performing a grid search over 4 different hyperparameter combinations using 5-fold cross-validation, what is the total number of models that will be trained during this evaluation process?

(A) 5

(B) 9

(C) 20

(D) 25

Q6. In standard k-fold cross-validation, what happens during the execution of the algorithm? (Select all that apply)

(A) The model is trained on $k-1$ folds of the data during each iteration

(B) The model is evaluated on 1 fold of the data that was excluded from training during that iteration

(C) The final model deployed to production is usually just the single model trained on the highest-scoring fold

(D) Each data point serves as part of the validation set exactly once across the entire k-fold process

Q7. When estimating the test error of a learning algorithm, how does LOOCV ($k = N$) generally compare to k-fold cross-validation with $k = 5$?

(A) LOOCV has higher bias and lower variance

(B) LOOCV has lower bias and higher variance

(C) LOOCV has lower bias and lower variance

(D) LOOCV has higher bias and higher variance

Q8. Why is 5-fold or 10-fold cross-validation generally preferred over 2-fold cross-validation in practice?

(A) 2-fold cross-validation requires training significantly more models, making it too slow

(B) In 2-fold cross-validation, the training sets are only half the size of the full dataset, which can introduce pessimistic bias (overestimating the model's true error rate)

(C) 2-fold cross-validation inherently causes train–test contamination

(D) 5-fold and 10-fold CV eliminate the need to keep a separate, untouched test set

Q9. Assume that as part of 4-fold cross-validation, the data was divided into parts f_1, f_2, f_3, f_4. Cross-validation was then performed. In this context, which of the following is True?

(A) In each iteration, one of f_1, f_2, f_3, f_4 will be used for training, and the remaining three parts will be used for validation

(B) In each iteration, three of f_1, f_2, f_3, f_4 will be used for training, and the remaining one part will be used for validation

(C) In each iteration, two of f_1, f_2, f_3, f_4 will be used for training, and the remaining two parts will be used for validation

(D) In each iteration, three of f_1, f_2, f_3, f_4 will be used for training, and the remaining one part will be used as the final held-out test set

Q10. Cross-validation is used to select the ridge penalty λ. Relative to the unregularized least-squares fit, the selected $\lambda > 0$ helps by:

(A) Reducing the trace of $(X^\top X + \lambda I)^{-1}$ relative to $(X^\top X)^{-1}$, and hence the total variance of the coefficient estimates

(B) Increasing the trace of $(X^\top X + \lambda I)^{-1}$, and hence the total variance of the coefficient estimates

(C) Reducing the bias of the coefficient estimates to zero

(D) Reducing the irreducible error σ^2

Q11. Which of the following is True about selecting the regularization parameter λ?

(A) λ should be chosen to minimize training set error

(B) λ should be chosen to minimize cross-validation error

(C) λ should be chosen to minimize test set error

(D) λ should always be set to zero when the training set is large

Q12. Which of the following statements are True? (Check all that apply)

(A) The performance of a learning algorithm on the training set will typically be better than its performance on the test set

(B) In k-fold cross validation, the learning sets across folds overlap substantially

(C) The test set should be used to tune hyperparameters

(D) Irreducible error places a lower bound on achievable expected test error

Q13. You tune hyperparameters with 5-fold cross-validation and then report the best cross-validation score as your estimate of generalization performance. Why is this estimate optimistically biased?

(A) Because 5 folds is too few to estimate the error accurately

(B) Because the same folds were used both to select among models and to report the score, so the winning score partly reflects a lucky fit to those folds

(C) Because cross-validation always underestimates error, regardless of tuning

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(D) Because the folds are not stratified

Q14. A dataset contains multiple records per patient. Which of the following are TRUE? (Select all that apply)

(A) A random row-wise split can place records from the same patient in both training and validation, inflating the score

(B) Splitting by patient ID (group k-fold) gives a more honest estimate of performance on unseen patients

(C) This is a form of leakage even though no target variable was used in preprocessing

(D) Increasing k fixes the problem without changing the splitting scheme

Q15. Why is stratified k-fold preferred over plain k-fold for a classification problem in which the positive class is 2% of the data?

(A) It reduces the number of models that must be trained

(B) It preserves the class proportion in every fold, preventing folds with very few or zero positives and reducing the variance of the estimate

(C) It guarantees the classifier will not overfit

(D) It eliminates the need for a separate test set

Q16. What is the purpose of *repeated* k-fold cross-validation (e.g. 10-fold repeated 5 times with different random partitions)?

(A) To reduce the bias of the error estimate

(B) To reduce the variance of the error estimate due to the particular random partition chosen

(C) To reduce the irreducible error

(D) To allow the test set to be reused safely

Q17. After k-fold cross-validation has been used to choose a hyperparameter setting, what is the standard practice for producing the final deployed model?

(A) Deploy the fold model with the best validation score

(B) Average the predictions of all k fold models and deploy that ensemble as the only option

(C) Refit the model with the chosen hyperparameters on the full training data

(D) Deploy the fold model with the median validation score

Q18. Which of the following statements about the cost of validation schemes are TRUE? (Select all that apply)

(A) A single holdout split requires one model fit

(B) k-fold cross-validation requires k fits per hyperparameter setting

(C) LOOCV on $N = 10{,}000$ points requires 10,000 fits, which is usually impractical for large models

(D) LOOCV is always cheaper than 10-fold cross-validation because each training set differs by only one point

Q19. Which of the following are TRUE about the relationship between the validation set and the test set? (Select all that apply)

(A) The validation set is used for model and hyperparameter selection; the test set is used once, for a final unbiased estimate

(B) Repeatedly checking the test set and adjusting the model turns it into a validation set and destroys its unbiasedness

(C) With k-fold cross-validation replacing the validation set, a separate held-out test set is still advisable

(D) Any model that scores well on the validation set is guaranteed to score equally well on the test set

Q20. There are two common ways to estimate classifier performance:
i) do n-fold cross validation (typically n is 5 or 10) or
ii) measure performance on a test set that is distinct from the learning and validation (if used) sets.

When will you use i) and when ii) and why?

(A) Use n-fold cross validation when the dataset is large, and a separate test set when it is small, since cross validation needs many examples per fold.

(B) Always use n-fold cross validation, since averaging over folds removes bias regardless of dataset size.

(C) Use n-fold cross validation when the dataset is small, since holding out a test set would shrink the learning set **L** further; use a separate test set when the dataset is large, since the folds' learning sets overlap heavily (60% common in 5-fold) and yield very similar models.

(D) Use a separate test set when the dataset is small to save computation, and n-fold cross validation when it is large so more folds can be built.

3 Regularization

Q1. What is the correct closed-form solution for the parameter vector $\hat{\beta}$ in ridge regression?

(A) $\hat{\beta} = (X^\top X)^{-1} X^\top y$

(B) $\hat{\beta} = (X^\top X + \lambda I)^{-1} X^\top y$

(C) $\hat{\beta} = (X^\top X - \lambda I)^{-1} X^\top y$

(D) $\hat{\beta} = (X^\top X + \lambda I)\, X^\top y$

Q2. Why does ridge regression guarantee a unique solution even when the number of features (p) is greater than the number of observations (N), or when features are perfectly collinear? (Select all that apply)

(A) Because ridge regression automatically drops redundant features to ensure $X^\top X$ is invertible

(B) Because adding λI (where $\lambda > 0$) to $X^\top X$ makes the resulting matrix strictly positive definite

(C) Because all eigenvalues of the matrix $(X^\top X + \lambda I)$ are strictly greater than zero

(D) Because the L_2 penalty converts the optimization problem into a non-convex space with a single global minimum

Q3. Which fundamental geometric property of the lasso (L_1) penalty leads it to produce sparse models (setting some coefficients exactly to zero)?

(A) The L_1 penalty function is continuously differentiable at the origin

(B) The L_1 penalty squares the coefficients, forcing smaller coefficients to shrink faster than larger ones

(C) The L_1 penalty's constraint region has sharp corners (vertices) on the axes, making it highly probable that the elliptical contours of the OLS loss function intersect the constraint region exactly on an axis

(D) Lasso adds a constant bias term to all coefficients, shifting many of them below zero

Q4. Consider the geometric interpretation of regularization in a 2D parameter space (β_1, β_2). Which of the following statements are TRUE? (Select all that apply)

(A) The constraint region for ridge regression is a circle/hypersphere

(B) The constraint region for lasso regression is a diamond/cross-polytope

(C) The unregularized OLS solution is always located exactly on the boundary of the regularization constraint regions

(D) If the unregularized OLS solution lies inside the constraint region, the penalized solution will be identical to the OLS solution

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Q5. In a ridge regression model, as the regularization hyperparameter λ approaches infinity ($\lambda \to \infty$), what happens to the estimated coefficients $\hat{\beta}$ and to the model's bias and variance?

(A) Coefficients approach ∞; bias approaches 0; variance approaches ∞

(B) Coefficients approach the OLS estimates; bias approaches 0; variance approaches 0

(C) Coefficients approach 0; bias approaches 0; variance approaches its maximum

(D) Coefficients approach 0; bias approaches its maximum; variance approaches 0

Q6. How do ridge and lasso regression typically behave when dealing with a group of highly correlated (collinear) features? (Select all that apply)

(A) Lasso tends to arbitrarily select one feature from the correlated group, assign it the full predictive weight, and shrink the others to zero

(B) Ridge tends to shrink the coefficients of highly correlated features towards each other, distributing the weight relatively evenly among them

(C) Both ridge and lasso perfectly preserve all correlated features with identical coefficient values

(D) Ridge regression will fail to converge when features are perfectly correlated

Q7. When applying ridge or lasso regression, why is it standard practice to standardize the features (mean 0, variance 1) and NOT penalize the intercept β_0? (Select all that apply)

(A) Penalizing the intercept would make the model's predictions dependent on the arbitrary origin/mean of the target variable y

(B) Standardizing features ensures that the regularization penalty is applied equitably based on predictive power, rather than being skewed by the arbitrary units of measurement of the features

(C) The intercept must be penalized to prevent the model from exhibiting infinite variance

(D) If features are not standardized, a feature with a very small numerical range (e.g. measured in km instead of mm) will have a naturally larger coefficient and thus be unfairly targeted by the penalty

Q8. The elastic net objective function combines both L_1 and L_2 penalties. Which specific limitation of lasso regression does elastic net primarily aim to solve?

(A) Lasso's inability to handle categorical variables

(B) Lasso's tendency to saturate and select at most n variables when the number of features p is strictly greater than the number of observations n

(C) Lasso's high computational cost compared to ridge regression

(D) Lasso's inability to perform variable selection

Q9. Increasing λ in ridge regression:

(A) Bias $\uparrow$, variance $\downarrow$

(B) Bias $\downarrow$, variance $\uparrow$

(C) Both $\downarrow$

(D) Both $\uparrow$

Q10. Suppose you have implemented a regularized linear regression model with λ set to a very large value. You observe high error on both the training set and the test set. Which of the following steps can be incorporated to lower the error on the testing dataset? (Select all that apply)

(A) Try decreasing the regularization parameter λ

(B) Try adding polynomial features

(C) Use fewer training examples

(D) Try using a smaller set of the features

Q11. The penalized form $\min_\beta \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$ and the constrained form $\min_\beta \|y - X\beta\|_2^2$ subject to $\|\beta\|_2^2 \leq t$ are related how?

(A) They are unrelated formulations that give different solution paths

(B) They are equivalent: for every $\lambda \geq 0$ there is a t giving the same solution, with t decreasing as λ increases

(C) They are equivalent only when X is orthonormal

(D) They are equivalent, with t increasing as λ increases

Q12. As λ is decreased from a very large value towards zero along the lasso regularization path, what is observed?

(A) Coefficients enter the model one at a time (or in groups), and the number of non-zero coefficients generally increases

(B) All coefficients become non-zero simultaneously at a single threshold value of λ

(C) The number of non-zero coefficients decreases monotonically

(D) Coefficients remain constant until $\lambda = 0$, at which point they jump to the OLS values

Q13. Which of the following are TRUE of the elastic net, with objective $\|y - X\beta\|_2^2 + \lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2$? (Select all that apply)

(A) Setting $\lambda_2 = 0$ recovers lasso, and setting $\lambda_1 = 0$ recovers ridge

(B) It exhibits a grouping effect: strongly correlated predictors tend to enter or leave the model together with similar coefficients

(C) It can produce sparse solutions, unlike pure ridge

(D) It can select at most n variables when $p > n$, exactly as lasso does

Q14. When $p > N$, what is true of the ordinary least squares problem, and why does this motivate regularization?

(A) $X^\top X$ is singular, so infinitely many β achieve zero training error; a penalty is needed to select a unique, stable solution

(B) $X^\top X$ is invertible but ill-conditioned, so OLS is unique though numerically unstable

(C) OLS has a unique solution with zero variance

(D) OLS is undefined because y cannot be projected onto the column space of X

Q15. Which of the following statements about scaling are TRUE? (Select all that apply)

(A) OLS coefficient estimates adapt exactly to a rescaling of a feature, so the fitted values are unchanged

(B) Ridge and lasso are not scale-invariant: rescaling a feature changes the effective strength of the penalty on it

(C) Standardization is therefore performed before fitting, using statistics computed on the training data only

(D) Because the penalty is applied to squared or absolute coefficients, scaling has no effect on which features are shrunk

Q16. For a finite $\lambda > 0$, ridge regression:

(A) Shrinks coefficients towards zero but generally does not set any of them exactly to zero

(B) Sets the least useful coefficients exactly to zero, performing variable selection

(C) Leaves coefficients unchanged but reduces the intercept

(D) Increases the magnitude of all coefficients

Q17. Assume that we obtain the weight vector $w = [4, 6, 8, 10]$ when $\lambda = 3$ is used in ridge regression. If λ is increased to 6, which of the following is most likely to be the updated weight vector?

(A) $w = [7, 9, 11, 13]$

(B) $w = [1, 3, 6, 9]$

(C) $w = [10, 12, 13, 15]$

(D) $w = [4, 6, 8, 10]$, unchanged

Q18. Assume that lasso is applied to a dataset containing only one feature. If λ is increased to a high value, which of the following is correct?

(A) The regression line may become horizontal

(B) The regression line may become vertical

(C) The regression line will remain the same

(D) The slope of the regression line will increase in magnitude

Q19. In ridge regression, if λ tends to zero, the solution approaches:

(A) Zero

(B) One

(C) Ordinary linear regression (OLS)

(D) Lasso regression

(E) Infinity